

NEMURO, Japan

WMO: 474200

Lat: 43.3315N

Lon: 145.5860E

Elev: 89

StdP: 14.65

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.3	15.8	1.6	6.0	17.8	4.1	6.8	18.8	39.3	27.6	34.8	25.2	10.8	240	0.709

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.4	76.4	69.3	73.1	67.8	70.5	66.4	71.1	74.4	69.3	71.9	67.4	69.7	11.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.0	110.5	73.1	68.2	104.1	71.1	66.4	97.5	69.4	35.1	74.9	33.6	72.1	32.0	69.8	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.8	27.7	24.5	9.3	83.6	3.6	4.1	6.7	86.5	4.6	88.9	2.6	91.2	0.0	94.2		
(5)			8.2	73.9	3.5	2.5	5.6	75.7	3.6	77.2	1.6	78.6	-0.9	80.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.1	25.2	24.5	30.4	38.8	46.3	52.2	59.5	64.0	61.8	52.9	41.8	30.7		
	DBStd	14.60	4.78	5.19	4.93	5.16	5.83	5.28	5.62	5.28	4.47	4.83	6.17	5.35		
	HDD50	3488	770	714	607	340	145	34	1	0	0	26	253	598		
	HDD65	7717	1235	1134	1072	787	578	384	190	85	118	375	696	1063		
	CDD50	1339	0	0	0	2	32	101	295	433	353	116	7	0		
	CDD65	93	0	0	0	0	0	1	18	52	22	0	0	0		
	CDH74	231	0	0	0	0	3	8	62	130	28	0	0	0		
	CDH80	31	0	0	0	0	0	2	8	20	1	0	0	0		
	WSAvg	11.9	13.6	12.6	12.6	12.4	11.9	10.1	9.0	9.5	10.7	12.7	13.7	13.8		
	PrecAvg	41.5	1.6	1.1	2.2	3.0	4.0	4.1	4.4	4.8	6.0	4.8	3.3	2.2		
	PrecMax	63.9	5.5	3.2	5.0	8.7	10.5	10.4	13.3	19.7	16.8	12.3	9.1	6.8		
(15) Precipitation	PrecMin	29.4	0.2	0.1	0.2	0.4	1.2	0.9	0.5	0.7	1.3	1.3	0.1	0.3		
	PrecStd	7.8	1.1	0.7	1.2	1.5	2.0	2.0	2.6	3.1	2.8	2.4	1.8	1.3		
	DB	39.6	40.0	47.4	61.7	69.2	72.9	80.1	82.6	77.6	68.0	58.7	48.7			
	MCWB	37.8	37.2	40.3	48.8	56.3	62.4	70.6	73.3	69.9	60.3	54.1	46.4			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	36.7	36.5	42.5	54.4	63.2	67.1	74.7	77.4	73.6	64.1	56.1	45.2			
	2% MCWB	34.5	33.9	38.3	45.3	53.6	59.8	68.6	70.1	68.3	58.6	52.4	42.9			
	5% DB	34.3	34.3	39.8	49.8	59.0	63.4	70.9	73.9	70.9	61.9	53.8	42.0			
	5% MCWB	31.9	31.8	36.1	43.4	51.4	58.1	66.5	68.6	66.5	57.5	50.6	39.6			
	10% DB	32.6	32.3	37.5	46.2	55.2	60.0	67.6	71.1	68.4	59.9	51.5	39.2			
	10% MCWB	30.2	30.1	34.3	41.5	49.6	56.2	64.6	67.7	64.5	55.9	48.1	36.4			
	0.4% WB	38.4	37.9	42.0	50.3	58.5	63.9	72.3	74.2	73.1	63.6	56.1	47.3			
	0.4% MCDB	39.4	39.6	45.2	59.5	64.7	69.8	78.0	80.9	75.5	65.3	57.3	48.4			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	35.0	34.5	39.1	46.6	54.9	60.9	69.6	71.6	70.0	60.5	53.2	43.2			
	2% MCDB	36.4	35.9	41.7	52.2	61.2	65.5	73.6	75.1	72.0	62.5	55.0	44.7			
	5% WB	32.5	32.3	36.8	44.3	52.6	58.7	67.1	70.0	67.7	58.5	51.0	39.8			
	5% MCDB	33.9	33.7	39.2	48.3	57.5	62.3	70.1	72.6	69.7	60.7	53.3	41.8			
	10% WB	30.6	30.4	34.7	42.3	50.5	56.6	64.9	68.3	65.3	56.8	48.7	36.8			
	10% MCDB	32.4	32.2	37.0	45.7	54.3	59.6	67.4	70.4	67.7	59.2	51.3	38.9			
	MDBR	8.8	9.5	9.4	11.0	11.8	11.0	10.6	9.4	9.0	9.7	10.0	9.1			
	5% DB	10.7	11.9	12.6	17.6	19.5	17.0	16.3	14.0	11.9	11.3	12.6	12.9			
(36) Mean Daily Temperature Range	5% DB	10.7	10.5	9.7	11.0	11.4	10.8	10.7	8.2	7.5	8.8	11.7	13.5			
	5% WB	10.9	10.9	12.0	16.0	17.5	15.5	15.5	12.4	10.5	9.8	12.1	12.7			
	MCWBR	11.4	10.4	10.2	10.9	11.3	10.6	10.6	8.3	7.8	9.0	12.3	13.8			
	MCWBR	11.4	10.4	10.2	10.9	11.3	10.6	10.6	8.3	7.8	9.0	12.3	13.8			
(40) Clear-Sky Solar Irradiance	taub	0.334	0.374	0.447	0.508	0.523	0.492	0.475	0.461	0.414	0.400	0.377	0.346			
	taud	2.220	2.086	1.943	1.860	1.910	2.097	2.216	2.273	2.353	2.244	2.205	2.191			
	Ebn at Noon	245	252	246	242	243	251	254	254	258	242	225	225			
	Edh at Noon	29	40	52	61	60	50	44	40	35	34	30	28			
(44) All-Sky Solar Radiation	RadAvg	551	830	1159	1433	1488	1476	1361	1237	1116	884	583	468			
	RadStd	42	45	85	117	132	136	121	134	94	51	46	40			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.58	N/A	N/A	N/A	N/A	N/A	N/A	-196	N/A	N/A
(47) Regional (4 neighbors)		+0.54	N/A	N/A	N/A	N/A	N/A	N/A	-149	N/A	N/A

Nomenclature: See separate page